

Project Activities by Phase

Specification phase – CIEMS

Chemical Injection Equipment Management System (CIEMS) | Business Analyst Challenge | Version 1.0

1. Approach

The specification is organized in **three phases**: (1) Requirements Elicitation, (2) Requirements Detailing, and (3) Validation & Handover. Durations are working-day estimates for a single Business Analyst with part-time support from UX, an architect, and the client team; they are meant as a planning baseline and assume stakeholders are available within 2 business days.

Timeline (estimate): Phase 1 $\approx$ 8 days · Phase 2 $\approx$ 14 days · Phase 3 $\approx$ 6 days $\rightarrow \approx$ **6 weeks** (including reviews and validation). Activities can overlap where dependencies allow.

2. Phase 1 – Requirements Elicitation

Objective: understand the business, the physical process, and the people, and collect needs, constraints, and open questions.

ID	Activity	Owner	Output	Effort	Depends on
1.1	Kick-off with client sponsor; confirm objectives, scope boundaries, and success criteria	BA, PM, Sponsor	Scope Statement (D01)	1 d	–
1.2	Identify and map stakeholders (Sales, Field Technicians, IT/Integration, Operations)	BA	Stakeholder map (D02)	1 d	1.1
1.3	Prepare elicitation plan and interview guides	BA	Elicitation plan (D03)	1 d	1.2
1.4	Analyze the process diagram (Site 1) and existing documents/spreadsheets	BA	Domain Notes & Initial Glossary	1 d	1.1
1.5	Interviews with Sales (registration, forecast in days, delivery planning)	BA	Interview notes (<i>supply forecasting, delivery planning, connectivity constraints</i>)	1.5 d	1.3
1.6	Interviews/shadowing with Field Technicians (field routines, connectivity, devices, browser limits)	BA, UX	Interview notes, personas (<i>supply forecasting, delivery planning, connectivity constraints</i>)	1.5 d	1.3
1.7	Technical session with IT / real-time system owners (tags, frequency, units, availability, security)	BA, Solutions Architect	Integration constraints	1 d	1.3
1.8	Collect volume/production sample data to validate the calculation logic (refills, gaps, zero production)	BA, Data Engineer	Sample dataset, data-quality findings	1.5 d	1.7
1.9	Elicitation workshop: consolidate needs, pains, and ideas (features, indicators)	BA, all	Consolidated needs list	0.5 d	1.5–1.7
1.10	Model AS-IS process (BPMN) and identify improvement points (<i>manual gaps and bottlenecks</i>)	BA	AS-IS model (D04)	1 d	1.5–1.6
1.11	Record assumptions, risks and open questions; agree on owners	BA	Assumptions & questions log (D06)	0.5 d	1.9

ID	Activity	Owner	Output	Effort	Depends on
1.12	Phase 1 review and go/no-go for detailing	BA, PM, Sponsor	Approved needs baseline	0.5 d	1.11

3. Phase 2 – Requirements Detailing

Objective: transform needs into precise, testable specifications and low-fidelity designs that development can estimate and build.

ID	Activity	Owner	Output	Effort	Depends on
2.1	Define actors, roles, and permission matrix (Sales, Field Technician, Admin)	BA	Permission matrix	1 d	1.12
2.2	Write functional requirements (registration, association, dashboard, alerts, export)	BA	FR list (D08)	2 d	2.1
2.3	Write non-functional requirements (browser-only, performance, security, availability, auditability)	BA, Solutions Architect	NFR list (D08)	1 d	1.7
2.4	Formalize business rules (max 3 products per well, AxC incompatibility, 20% threshold, allocation)	BA	Business rules catalog (D08)	1.5 d	2.2
2.5	Specify calculations (injection rate, per-well consumption, autonomy) with worked examples and edge cases	BA, Data Engineer	Calculation spec (D13)	2 d	1.8, 2.4
2.6	Model domain/data (ER) and validate with architect	BA, Solutions Architect	Data model (D10)	1 d	2.2
2.7	Write use cases and TO-BE flows (UML / BPMN)	BA	Use cases, diagrams (D09)	1.5 d	2.2
2.8	Write user stories with acceptance criteria (Given/When/Then)	BA	Story set (D08)	2 d	2.2, 2.4
2.9	Specify the integration contract with the real-time data system	BA, Integration Eng.	Integration spec (D11)	1.5 d	1.7
2.10	Design lo-fi prototype and navigation flow; run internal walkthrough	BA, UX	Lo-fi prototype (D12)	2 d	2.2, 2.7
2.11	Define KPIs, dashboards, and suggested features with justification	BA, Product	KPI spec (D14)	1 d	2.2
2.12	Build traceability matrix (need → requirement → rule → story → screen)	BA	RTM (D16)	1 d	2.8, 2.10
2.13	Prioritize backlog (MoSCoW + WSJF) and propose release plan; recommend methodology	BA, PO	Backlog & methodology (D15)	1 d	2.8
2.14	Internal quality review (peer review of consistency, ambiguity, testability)	BA, Peer BA	Review log	0.5 d	2.12

4. Phase 3 – Validation & Handover

Objective: obtain client agreement and prepare a backlog that is ready for the first sprint.

ID	Activity	Owner	Output	Effort	Depends on
3.1	Prototype usability sessions with 2–3 Sales users and 2–3 Field Technicians	BA, UX	Feedback list	1.5 d	2.10

ID	Activity	Owner	Output	Effort	Depends on
3.2	Requirements review workshop with the client (walkthrough by use case)	BA, Client	Change requests, decisions	1 d	2.14
3.3	Resolve open questions and update documents/prototype	BA, UX	Updated baseline v1.1	1.5 d	3.1, 3.2
3.4	Draft UAT scenarios from acceptance criteria	BA, QA	UAT scenarios (D17)	1 d	3.3
3.5	Formal sign-off of the specification	Sponsor, BA	Signed approval (D18)	0.5 d	3.3
3.6	Load backlog into Jira / Azure Boards; define DoR / DoD; refine first sprint with the dev team	BA, PO, Dev lead	Ready backlog (D19)	1 d	3.5

5. Key risks affecting the plan

Risk	Impact	Mitigation
Late access to real-time system owners	Integration spec delayed	Schedule IT session in week 1; use sample data files as fallback
Calculation rules not agreed by Sales (allocation criteria)	Rework of KPIs and dashboard	Validate worked examples with Sales before writing stories
Field technicians unavailable for testing	Poor usability	Remote sessions, short 30-minute slots, prototype on tablet
Scope growth from new site-specific needs	Schedule slip	Change-control log; site-specific fields handled as configuration
Intermittent field connectivity impacting real-time browser usability	Field Technicians unable to complete tank-well associations on-site.	Design client-side form validation and clear feedback indicators for network status within browser UI.